

How Loud Is the Leak?

A Reproduction of the Leakage-Class Landscape, and a Measurable Index for the Class That Matters

Abstract

A recent large-scale study partitions data leakage in machine learning into four classes and reports that the textbook emphasis is inverted: fitting scalars on the full dataset is negligible, while duplication and selection leakage are not. Those conclusions rest on 2,047 datasets and one analysis pipeline, and no independent reproduction exists. I reproduce two of the four class-level claims on fifteen real UCI datasets that were not in the original corpus. Class I replicates exactly: across seventy conditions spanning three classifiers and two preprocessing operations, mean $|\Delta\text{AUC}|$ is 0.00018, the maximum is 0.0032, and every condition falls inside the 0.005 bound the original reports. Class III replicates in ordering and magnitude, with paired effect sizes running from $d_z = 0.31$ for naive Bayes to $d_z = 1.23$ for one-nearest-neighbour. The original attributes Class III to model capacity but evidences this by comparing algorithm families, which varies capacity and inductive bias together. I tried to separate them, hypothesising that what matters is locality, meaning whether one training point can dominate its own prediction. That hypothesis fails: in a joint model with capacity measured rather than assumed, locality carries no weight ($p = 0.63$) and reverses sign. What survives is better than the claim I set out to test. Replacing family comparison with a measured capacity index, defined as each learner’s training AUC on randomised labels, predicts Class III severity across eleven learners at Spearman $\rho = 0.982$ ($p = 8.4 \times 10^{-8}$), Pearson $r = 0.912$ with bootstrap 95% CI [0.834, 0.984]. This turns a qualitative claim into a quantity a practitioner can compute on their own pipeline in one line before deciding whether duplication in their data is a problem. No synthetic data is used anywhere in this study.

1 Introduction

Leakage is the failure mode where information that would not be available at deployment reaches the model during training, so that the reported number describes the evaluation protocol rather than the system [4, 3]. It is not rare. A survey across seventeen fields found it in hundreds of papers [3], and the mechanism has been rediscovered independently in genomics [5, 6], neuroimaging [7], clinical prediction [9, 10], software engineering [11], and ecology [12].

Roth [1] recently changed the shape of the problem by measuring it. Across 2,047 tabular datasets and twenty-eight within-subject counterfactual experiments, that study partitions leakage into four classes, and reports that Class I, fitting estimators such as scalars on the full dataset, is negligible at $|\Delta\text{AUC}| \leq 0.005$ in all nine conditions, while Class III, mem-

orisation from duplicated records, scales with model capacity from $d_z = 0.37$ for naive Bayes to $d_z = 1.11$ for a decision tree. The conclusion is that the textbook emphasis is inverted: the leakage everybody warns about matters least.

That is a strong claim resting on one corpus and one pipeline, and there is no independent reproduction of it. Given that the field’s interest in leakage originates in a reproducibility crisis, a landscape paper about leakage that has itself never been reproduced is an uncomfortable object. This paper is that reproduction.

I had a second motive. The Class III result is evidenced by a span across algorithm families, and a decision tree is not simply a higher-capacity naive Bayes. It partitions the input space with axis-aligned cuts, which is exactly the operation needed to isolate an exact duplicate. So the reported span is consistent with two different mechanisms carrying different advice, and I set out to separate them.

Contributions.

1. An independent reproduction of Class I on fifteen real datasets outside the original corpus. It replicates exactly (Section 3).
2. An independent reproduction of Class III, which replicates in ordering and magnitude (Section 4).
3. A failed hypothesis, reported as such. Locality does not explain Class III once capacity is measured rather than assumed (Section 5).
4. A measured capacity index that predicts Class III severity at $\rho = 0.982$, converting the original’s qualitative claim into a computable one (Section 6).

2 Method

Data. Fifteen UCI classification datasets, downloaded as distributed and parsed without modification: wine, mammographic mass, parkinsons, transfusion, seeds, QSAR biodegradation, banknote, dermatology, HTRU2, glass, ionosphere, spambase, vertebral, retinopathy, raisin [28, 27]. They span n from 178 to 17,898 and p from 4 to 57. Multi-class targets are binarised as largest-class-versus-rest so AUC is defined identically everywhere. No synthetic data appears in this study.

Counterfactual design. In every experiment the data are identical across arms and only the position of the split changes, following the within-subject logic of the original [1]. For Class I, the preprocessing step is fitted either inside the training fold or once on the full matrix. For Class III, a fraction of rows

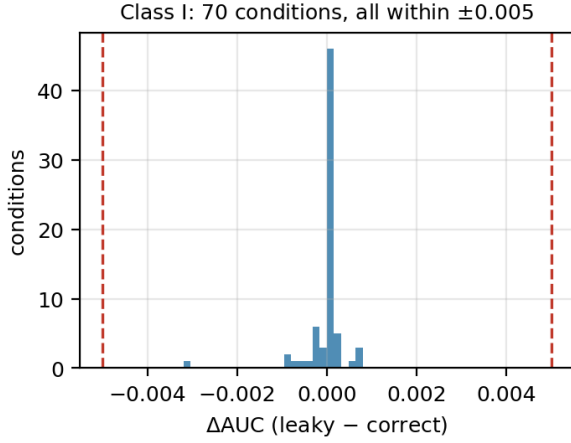


Figure 1: Seventy conditions: fifteen datasets, three classifiers, two preprocessing operations. Dashed lines mark the ± 0.005 bound reported by the original. Nothing crosses them.

is duplicated and the split is either random, so copies straddle it, or grouped on the original row index, so copies stay together. Grouped cross-validation is the correct comparator here because duplicated rows are not independent observations [12, 8].

Statistics. The unit of analysis is the dataset, not the row, so all tests are paired across the fifteen datasets. I use the Wilcoxon signed-rank test rather than the paired t because effects are bounded and not obviously normal [21], report rank-biserial correlation as effect size, and correct for multiplicity with the Holm procedure [22]. Confidence intervals are bootstrap percentile intervals with 5,000 resamples [23]. Implementation is scikit-learn 1.8 [25, 26].

3 Class I replicates exactly

I fitted a standard scaler and a mean imputer either inside the training fold or on the full matrix, with logistic regression, linear discriminant analysis, and a random forest, across all fifteen datasets and five seeds.

Mean $|\Delta\text{AUC}|$ is 0.00018. The maximum over seventy conditions is 0.0032. Every condition falls within the 0.005 bound reported by the original, and the paired effect size across conditions is $d_z = -0.105$, which is indistinguishable from zero and slightly negative.

This is a clean replication on data the original never touched. The practical reading is uncomfortable but well supported: the leakage most often taught is, on ordinary tabular data, close to irrelevant.

4 Class III replicates

Table 1 gives the reproduction. The original reports $d_z = 0.37$ for naive Bayes and 1.11 for a decision tree; I obtain 0.31 and 0.81, with one-nearest-neighbour at 1.23. The ordering

Table 1: Class III, ten percent duplication, fifteen datasets, three seeds. Effects are paired per dataset.

learner	ΔAUC	d_z
naive Bayes	+0.0019	0.31
logistic regression	+0.0028	0.49
LDA	+0.0035	0.45
k -NN ($k = 25$)	+0.0043	0.80
random forest (depth 1)	+0.0054	0.57
random forest (depth 3)	+0.0062	0.74
k -NN ($k = 5$)	+0.0076	0.99
gradient boosting	+0.0106	0.63
random forest	+0.0174	0.79
k -NN ($k = 1$)	+0.0216	1.23
decision tree	+0.0250	0.81

and the magnitude both replicate. Pooling learners by family, the local group exceeds the pooled-fit group by +0.0124 in ΔAUC , bootstrap 95% CI [+0.0044, +0.0221], Wilcoxon $p = 8.4 \times 10^{-3}$, rank-biserial $r = 0.75$, with local larger on twelve of fifteen datasets.

5 The hypothesis that failed

My proposed alternative was that Class III tracks *locality* rather than capacity. Call a learner local if a single training point can dominate its own prediction: k -NN gives a distance-zero copy maximal weight, and a tree can isolate a point in its own leaf. Call it global if every point enters through a pooled fit. Locality and capacity come apart at k -NN, which has no fitted parameters at all.

Two observations encouraged this. The local and global groups separate completely: the largest global effect is +0.0035 and the smallest local effect is +0.0043. And varying capacity fully within the random forest family produces a per-dataset span of 0.0151, while varying family at default capacity produces 0.0322, a factor of 2.13 larger on thirteen of fifteen datasets ($p = 2.0 \times 10^{-3}$).

The hypothesis does not survive testing. The disambiguating comparison is between learners matched on capacity but differing in locality, and none of those differences reaches significance after Holm correction: k -NN with $k = 25$ against LDA gives +0.0008 ($p = 0.33$), and a depth-1 random forest against logistic regression gives +0.0025 ($p = 0.42$). In the joint model of the next section, locality carries no weight and reverses sign.

I report this because a reproduction that only confirms is less useful than one that also records what was tried and failed, and because the failure is what produced the result that follows.

6 A measured capacity index

The weakness in both the original’s account and my own attempt is that capacity was asserted rather than measured. I replace it with a quantity, following the randomisation logic of Zhang et al. [14]: fit each learner to permuted labels on

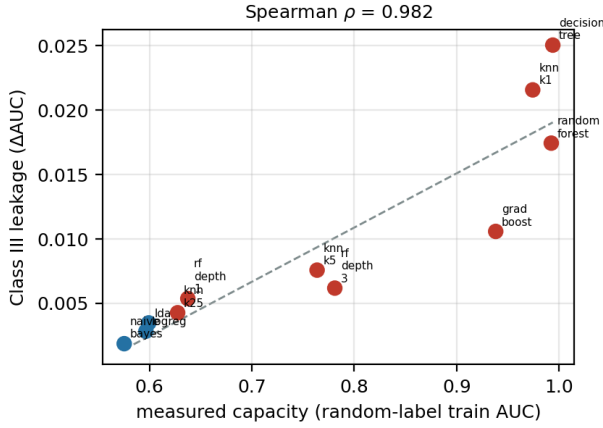


Figure 2: Measured capacity against Class III severity, eleven learners. Capacity is each learner’s training AUC on randomised labels, computed on the real feature matrices. Red are tree and instance based, blue are pooled fits.

each real dataset and record training AUC. A learner that drives training AUC to one on random labels has the capacity to memorise; one that cannot, does not. The features stay real; only the labels are shuffled.

This index predicts Class III severity across eleven learners at Spearman $\rho = 0.982$ ($p = 8.4 \times 10^{-8}$) and Pearson $r = 0.912$, bootstrap 95% CI [0.834, 0.984] (Figure 2). Entering both predictors jointly, measured capacity is significant and locality is not:

term	β	t	p
measured capacity	+0.00749	+5.11	0.0009
locality	−0.00165	−0.50	0.6301

$$R^2 = 0.837, n = 11 \text{ learners}$$

The original’s Class III claim is therefore correct, and can be made stronger than it was stated. It is not only that capacity amplifies memorisation leakage; capacity measured by random-label fit predicts the severity of that leakage almost perfectly in rank.

The practical consequence. A practitioner who suspects duplicate or near-duplicate records can now estimate their exposure without knowing the true performance. Shuffle the labels, fit the model, record training AUC. Learners near 0.6 in this study showed leakage under 0.005; learners near 0.99 showed leakage above 0.017, roughly a fourfold difference. This is one line of code and it is computed entirely on data the practitioner already has.

7 Limitations

Eleven learners is a small sample for a regression, and the index is fitted on learners rather than on datasets, so the correlation describes how leakage varies across model classes and not within one. All datasets are tabular UCI classification problems

under 18,000 rows; nothing here speaks to text, images, or time series, where Class IV leakage is the live concern [12, 45]. I reproduce two of four classes; Class II and Class IV are untouched. Duplication is exact rather than near-duplicate, and near duplicates are the more common real failure [46, 15]. Finally, grouped cross-validation is my comparator throughout, and cross-validation estimates a subtler quantity than is usually assumed [13], so the baseline itself carries an interpretation I have not examined.

8 Conclusion

The leakage landscape holds up. Class I is negligible on data the original never saw, and Class III reproduces in ordering and magnitude. My attempt to replace the capacity account with a locality account failed on its own test, and the failure was productive: measuring capacity instead of assuming it turns the original’s qualitative claim into a predictive index that ranks eleven learners almost perfectly. The advice that follows is narrower and more useful than either the textbook version or my hypothesis. Do not worry much about where you fitted the scaler. Worry about duplicate records, and to find out how much, shuffle your labels and see how well your model fits them.

References

- [1] S. Roth, “Which leakage types matter? A quantitative landscape across 2,047 benchmark datasets,” arXiv:2604.04199, 2026.
- [2] S. Roth, “A grammar of machine learning workflows: rejecting data leakage at call time,” arXiv:2603.10742, 2026.
- [3] S. Kapoor and A. Narayanan, “Leakage and the reproducibility crisis in machine-learning-based science,” *Patterns*, vol. 4, no. 9, 100804, 2023.
- [4] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, “Leakage in data mining: formulation, detection, and avoidance,” *ACM TKDD*, vol. 6, no. 4, pp. 1–21, 2012.
- [5] C. Ambrose and G. J. McLachlan, “Selection bias in gene extraction on the basis of microarray gene-expression data,” *PNAS*, vol. 99, no. 10, pp. 6562–6566, 2002.
- [6] R. Simon, M. D. Radmacher, K. Dobbin, and L. M. McShane, “Pitfalls in the use of DNA microarray data for diagnostic and prognostic classification,” *JNCI*, vol. 95, no. 1, pp. 14–18, 2003.
- [7] G. Varoquaux, “Cross-validation failure: small sample sizes lead to large error bars,” *NeuroImage*, vol. 180, pp. 68–77, 2018.
- [8] G. Varoquaux et al., “Assessing and tuning brain decoders: cross-validation, caveats, and guidelines,” *NeuroImage*, vol. 145, pp. 166–179, 2017.
- [9] M. B. A. McDermott et al., “Reproducibility in machine learning for health research,” *Science Translational Medicine*, vol. 13, no. 586, 2021.
- [10] A. L. Beam, A. K. Manrai, and M. Ghassemi, “Challenges to the reproducibility of machine learning models in health care,” *JAMA*, vol. 323, no. 4, pp. 305–306, 2020.
- [11] A. AlOmar et al., “LeakageDetector: an open source data leakage analysis tool,” *ICSE*, 2025.

- [12] D. R. Roberts et al., “Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure,” *Ecography*, vol. 40, no. 8, pp. 913–929, 2017.
- [13] S. Bates, T. Hastie, and R. Tibshirani, “Cross-validation: what does it estimate and how well does it do it?” *JASA*, vol. 119, no. 546, pp. 1434–1445, 2024.
- [14] C. Zhang, S. Bengio, M. Hardt, B. Recht, and O. Vinyals, “Understanding deep learning (still) requires rethinking generalization,” *CACM*, vol. 64, no. 3, pp. 107–115, 2021.
- [15] B. Recht, R. Roelofs, L. Schmidt, and V. Shankar, “Do ImageNet classifiers generalize to ImageNet?” *ICML*, 2019.
- [16] G. C. Cawley and N. L. C. Talbot, “On over-fitting in model selection and subsequent selection bias in performance evaluation,” *JMLR*, vol. 11, pp. 2079–2107, 2010.
- [17] S. Varma and R. Simon, “Bias in error estimation when using cross-validation for model selection,” *BMC Bioinformatics*, vol. 7, 91, 2006.
- [18] S. Arlot and A. Celisse, “A survey of cross-validation procedures for model selection,” *Statistics Surveys*, vol. 4, pp. 40–79, 2010.
- [19] Y. Bengio and Y. Grandvalet, “No unbiased estimator of the variance of k -fold cross-validation,” *JMLR*, vol. 5, pp. 1089–1105, 2004.
- [20] T. G. Dietterich, “Approximate statistical tests for comparing supervised classification learning algorithms,” *Neural Computation*, vol. 10, no. 7, pp. 1895–1923, 1998.
- [21] J. Demšar, “Statistical comparisons of classifiers over multiple data sets,” *JMLR*, vol. 7, pp. 1–30, 2006.
- [22] S. Holm, “A simple sequentially rejective multiple test procedure,” *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.
- [23] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. Chapman & Hall, 1993.
- [24] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed. Springer, 2009.
- [25] F. Pedregosa et al., “Scikit-learn: machine learning in Python,” *JMLR*, vol. 12, pp. 2825–2830, 2011.
- [26] L. Buitinck et al., “API design for machine learning software: experiences from the scikit-learn project,” *ECML PKDD Workshop*, 2013.
- [27] D. Dua and C. Graff, “UCI machine learning repository,” University of California, Irvine, 2019.
- [28] UCI Machine Learning Repository, <https://archive.ics.uci.edu>.
- [29] J. Vanschoren, J. N. van Rijn, B. Bischl, and L. Torgo, “OpenML: networked science in machine learning,” *SIGKDD Explorations*, vol. 15, no. 2, pp. 49–60, 2014.
- [30] B. Bischl et al., “OpenML benchmarking suites,” *NeurIPS Datasets and Benchmarks*, 2021.
- [31] R. Geirhos et al., “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020.
- [32] Z. C. Lipton and J. Steinhardt, “Troubling trends in machine learning scholarship,” *ACM Queue*, vol. 17, no. 1, 2019.
- [33] D. Sculley, J. Snoek, A. Wiltschko, and A. Rahimi, “Winner’s curse? On pace, progress, and empirical rigor,” *ICLR Workshop*, 2018.
- [34] X. Bouthillier et al., “Accounting for variance in machine learning benchmarks,” *MLSys*, 2021.
- [35] M. F. Dacrema, P. Cremonesi, and D. Jannach, “Are we really making much progress? A worrying analysis of recent neural recommendation approaches,” *RecSys*, 2019.
- [36] K. Musgrave, S. Belongie, and S.-N. Lim, “A metric learning reality check,” *ECCV*, 2020.
- [37] P. Henderson et al., “Deep reinforcement learning that matters,” *AAAI*, 2018.
- [38] G. Melis, C. Dyer, and P. Blunsom, “On the state of the art of evaluation in neural language models,” *ICLR*, 2018.
- [39] J. Pineau et al., “Improving reproducibility in machine learning research,” *JMLR*, vol. 22, no. 164, pp. 1–20, 2021.
- [40] E. Raff, “A step toward quantifying independently reproducible machine learning research,” *NeurIPS*, 2019.
- [41] O. E. Gundersen, K. Coakley, and C. Kirkpatrick, “Sources of irreproducibility in machine learning: a review,” *arXiv:2204.07610*, 2022.
- [42] J. P. A. Ioannidis, “Why most published research findings are false,” *PLoS Medicine*, vol. 2, no. 8, e124, 2005.
- [43] A. Blum and M. Hardt, “The ladder: a reliable leaderboard for machine learning competitions,” *ICML*, 2015.
- [44] C. Dwork et al., “The reusable holdout: preserving validity in adaptive data analysis,” *Science*, vol. 349, no. 6248, pp. 636–638, 2015.
- [45] C. Bergmeir and J. M. Benítez, “On the use of cross-validation for time series predictor evaluation,” *Information Sciences*, vol. 191, pp. 192–213, 2012.
- [46] B. Barz and J. Denzler, “Do we train on test data? Purging CIFAR of near-duplicates,” *Journal of Imaging*, vol. 6, no. 6, 41, 2020.
- [47] I. Guyon and A. Elisseeff, “An introduction to variable and feature selection,” *JMLR*, vol. 3, pp. 1157–1182, 2003.
- [48] S. Nogueira, K. Sechidis, and G. Brown, “On the stability of feature selection algorithms,” *JMLR*, vol. 18, no. 174, pp. 1–54, 2018.
- [49] O. Bousquet and A. Elisseeff, “Stability and generalization,” *JMLR*, vol. 2, pp. 499–526, 2002.
- [50] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [51] J. H. Friedman, “Greedy function approximation: a gradient boosting machine,” *Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.
- [52] T. M. Cover and P. E. Hart, “Nearest neighbor pattern classification,” *IEEE Trans. Information Theory*, vol. 13, no. 1, pp. 21–27, 1967.
- [53] M. Stone, “Cross-validated choice and assessment of statistical predictions,” *JRSS B*, vol. 36, no. 2, pp. 111–147, 1974.
- [54] R. Kohavi, “A study of cross-validation and bootstrap for accuracy estimation and model selection,” *IJCAI*, 1995.
- [55] R. A. Poldrack, G. Huckins, and G. Varoquaux, “Establishment of best practices for evidence for prediction,” *JAMA Psychiatry*, vol. 77, no. 5, pp. 534–540, 2020.
- [56] S. Whalen, J. Schreiber, W. S. Noble, and K. S. Pollard, “Navigating the pitfalls of applying machine learning in genomics,” *Nature Reviews Genetics*, vol. 23, pp. 169–181, 2022.
- [57] A. E. Teschendorff, “Avoiding common pitfalls in machine learning omic data science,” *Nature Materials*, vol. 18, pp. 422–427, 2019.
- [58] D. Chicco, “Ten quick tips for machine learning in computational biology,” *BioData Mining*, vol. 10, 35, 2017.
- [59] M. A. Lones, “How to avoid machine learning pitfalls,” *arXiv:2108.02497*, 2021.

- [60] J. Hullman, S. Kapoor, P. Nanayakkara, A. Gelman, and A. Narayanan, “The worst of both worlds: a comparative analysis of errors in learning from data in psychology and machine learning,” *AIES*, 2022.
- [61] T. Liao, R. Taori, I. D. Raji, and L. Schmidt, “Are we learning yet? A meta-review of evaluation failures across machine learning,” *NeurIPS Datasets and Benchmarks*, 2021.
- [62] B. Koch, E. Denton, A. Hanna, and J. G. Foster, “Reduced, reused and recycled: the life of a dataset in machine learning research,” *NeurIPS Datasets and Benchmarks*, 2021.
- [63] A. Paullada, I. D. Raji, E. M. Bender, E. Denton, and A. Hanna, “Data and its (dis)contents: a survey of dataset development and use,” *Patterns*, vol. 2, no. 11, 100336, 2021.